

CO2

ASSESSMENT TOOL 2



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COURSE

CRYPTOGRAPHY AND NETWORK SECURITY



COURSE CODE

CSA5113

1. Secure Data Transmission using Block Cipher

A. Explain why this issue occurs.

When the company uses **Electronic Codebook (ECB) mode**, each plaintext block is encrypted independently using the same encryption key. As a result, if two plaintext blocks contain the same data, they produce exactly the same ciphertext block. Since financial records often contain repeated information, repeated ciphertext patterns become visible. An attacker may not know the actual data, but they can identify repeated patterns and make assumptions about the transmitted information. Therefore, ECB mode is not suitable for transmitting sensitive financial data.

B. Suggest a more secure block cipher mode and justify your choice.

A better choice is **Cipher Block Chaining (CBC) mode**.

In CBC mode, every plaintext block is first combined with the previous ciphertext block before encryption. The first block uses an **Initialization Vector (IV)**. Because each block depends on the previous ciphertext, identical plaintext blocks produce different ciphertext blocks.

Justification:

- Hides repeated patterns in the plaintext.
- Provides better confidentiality than ECB.
- Makes it difficult for attackers to identify repeated financial records.
- Widely used for secure transmission of sensitive information.

Conclusion: CBC mode is more secure than ECB because it prevents pattern leakage and provides stronger protection for financial data.

2. Choosing an Encryption Algorithm for Banking System

A. Identify the most suitable algorithm.

Among **DES**, **3-DES**, and **Blowfish**, the most suitable algorithm for the banking application is **Blowfish**.

B. Justify your choice based on security strength and efficiency.

A banking system requires both strong security and good performance.

- **DES** uses a **56-bit key**, which is no longer considered secure because it can be broken using brute-force attacks.
- **3-DES** improves security by applying DES three times with a **168-bit key**, but it is much slower due to multiple encryption operations.

- **Blowfish** supports keys up to **448 bits**, provides strong encryption, and performs faster than both DES and 3-DES.

Justification:

- Strong protection against attacks.
- Faster encryption and decryption.
- Suitable for handling large numbers of banking transactions.
- Reduces processing time while maintaining high security.

Conclusion: Blowfish is the best choice because it offers an excellent balance between security and performance.

3. Key Exchange in an Unsecured Network

A. Explain how the key exchange works.

Diffie-Hellman allows two users to establish a common secret key without sending the secret key directly over the network.

The process is as follows:

1. Both users agree on two public values: a prime number (**p**) and a generator (**g**).
2. User A selects a private key and computes a public key.
3. User B also selects a private key and computes another public key.
4. They exchange only their public keys.
5. Each user combines the received public key with their own private key to calculate the same shared secret key.
6. This shared secret is then used for secure communication.

Since only public keys are transmitted, the actual secret key is never exposed on the network.

B. Identify a possible security threat and suggest a solution.

The major security threat is the **Man-in-the-Middle (MITM) attack**.

In this attack, an attacker intercepts the exchanged public keys and replaces them with fake keys. Both users unknowingly establish separate secret keys with the attacker instead of with each other. The attacker can then read, modify, or forward all messages.

Solution:

- Use **Digital Certificates** to verify the identity of users.
- Use **Digital Signatures** to authenticate exchanged public keys.
- Implement secure protocols such as **TLS/SSL**, which provide authentication before the key exchange.

Conclusion: Authentication mechanisms prevent attackers from replacing public keys and ensure secure communication.

4. Public Key Cryptography for Email Security

A. Describe how RSA can be used for encryption and decryption.

RSA uses two different keys:

- **Public Key** – shared with everyone.
- **Private Key** – kept secret by the owner.

When a sender wants to send a confidential email:

1. The sender encrypts the email using the receiver's **public key**.
2. The encrypted email is transmitted over the internet.
3. Only the receiver, who possesses the corresponding **private key**, can decrypt and read the message.

This ensures that even if someone intercepts the email, they cannot understand its contents without the private key.

B. Explain the importance of key distribution and management.

Proper key management is essential for maintaining RSA security.

Key management involves:

- Generating strong public and private keys.
- Securely storing private keys.
- Safely distributing public keys.
- Periodically replacing old keys.

- Revoking compromised keys.

If a private key is leaked or stolen, attackers can decrypt confidential emails. Similarly, if users receive fake public keys, they may unknowingly send encrypted emails to attackers.

Conclusion: Effective key distribution and management ensure confidentiality, authentication, and secure email communication.

5. Cryptography for IoT Devices

A. Analyze the constraints of IoT devices.

IoT devices, such as smart home sensors and wearable devices, have limited hardware resources.

Their main constraints include:

- Limited processing power.
- Low memory capacity.
- Limited storage space.
- Battery-powered operation requiring low energy consumption.
- Need for fast and lightweight encryption algorithms.

Because of these limitations, heavy cryptographic algorithms may reduce performance and battery life.

B. Recommend a suitable cryptographic approach and justify your answer.

The most suitable cryptographic approach is **Elliptic Curve Cryptography (ECC)**.

ECC provides the same level of security as RSA while using much smaller key sizes. This reduces computation time, memory usage, and energy consumption.

Justification:

- Smaller key size provides equivalent security.
- Faster encryption and decryption on low-power devices.
- Lower memory and storage requirements.

Conclusion: ECC is the preferred cryptographic solution for smart home IoT devices because it delivers strong security while efficiently utilizing limited hardware resources.